



Figure 3: WebADM Account



Figure 4: OpenOTP setting

```
gunzip webadm-all-in-one-1.1.x.sh.gz
sh webadm-all-in-one-1.1.x.sh
```

For a basic installation, just keep hitting Enter till the installation is over. My installation directory is `/opt/webadm`. This script creates the WebADM system user, sets filesystem permissions, creates directories for configuration files, logs, etc.

Configuring OpenLDAP

The steps below are if you are setting up a new LDAP server. If you already have an existing OpenLDAP installation, then edit the configuration files `/opt/webadm/conf/webadm.conf` and `/opt/webadm/conf/servers.xml` to make the necessary changes. These files are self-explanatory and well-documented, but you can scan the procedure below as well.

Make sure that OpenLDAP related packages are installed, using your distribution's package manager. Copy the WebADM schema to the existing schema directory, and add this schema to the `/etc/openldap/slapd.conf` file:

```
cp /opt/webadm/doc/OpenLDAP.schema /etc/openldap/schema/webadm.schema
include /etc/openldap/schema/webadm.schema
```

Now we have to modify our suffix and `rootdn`, and also add ACLs for the webadm user. You could instead use the default LDAP user for webadm too, but it's not recommended, for security reasons. Here, I'm using `example.com` as my DC:

```
Suffix "dc=example, dc=com"
Rootdn "cn=Manager, dc=example, dc=com"
```

Add admin ACLs for the WebADM user (read and write permission to the entire LDAP tree) in `slapd.conf`:

```
Access to *
by dn="cn=admin, dc=example, dc=com" write
by self write
```

Now, we add the domain to the OpenLDAP tree. Create an LDIF file, with your favourite editor, and save it as `example.ldif`, with these contents (modify them to suit your domain):

```
dn : dc=example, dc=com
dc: example
ou: rootObject
objectClass: top
objectClass: dcObject
objectClass: organizationalUnit
dn: cn=admin, dc=example, dc=com
cn: admin
sn: admin
objectClass: person
objectClass: inetOrgPerson
```

Initialise the OpenLDAP directory by adding the just-created LDIF:

```
# slapadd -v -l example.ldif
```



Note: If you have an existing directory, then use `"ldapadd"` to add this `"admin"` user instead of `slapadd`.

Finally, restart OpenLDAP with `/etc/init.d/ldap restart`.

Configuring MySQL

We must create a `webadm` database to store audit logs and localised messages, and a `webadm` user with a password of our choice, with full permissions on that database. My example uses the username `"webadm"` and the password `"password"`. Make sure the MySQL service is running; log in with the client, as `root`; create the `webadm` database, and assign permissions, as follows:

```
mysql -u root -p
mysql> create database webadm;
```